

Balanced Mirror Sums

Problem Statement

You are given an even integer n and an array $a_1, a_2, \dots, a_n$ of integers.

For each i ($1 \leq i \leq n/2$), define the *mirrored pair sum*:

$$p_i = a_i + a_{n-i+1}.$$

An array is called **balanced** if the sequence $p_1, p_2, \dots, p_{n/2}$ is nondecreasing. Formally,

$$p_1 \leq p_2 \leq \dots \leq p_{n/2}.$$

You are allowed to perform at most one operation: choose any two indices i and j ($1 \leq i, j \leq n$, $i \neq j$) and swap a_i and a_j .

Determine whether it is possible to make the array balanced using at most one swap.

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

Each test case consists of two lines:

- The first line contains two integers n and k ($2 \leq n \leq 2 \cdot 10^5$, n is even, $0 \leq k \leq 1$). For this problem, k will always be 1.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of the array.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single line:

- YES — if it is possible to make the array balanced using at most one swap.
- NO — otherwise.

Example

Input:

```

4
6 1
5 6 9 8 1 7
6 0
5 6 9 8 7 1
6 0
5 6 9 8 1 7
4 1
10 1 2 9

```

Output:

```

YES
YES
NO
YES

```

Explanation

- **Test case 1:** The mirrored pair sums are $5 + 7 = 12$, $6 + 1 = 7$, $9 + 8 = 17$. Not nondecreasing. By swapping elements 1 and 7, the array becomes $[5, 6, 9, 8, 7, 1]$, with sums $5 + 1 = 6$, $6 + 7 = 13$, $9 + 8 = 17$, which is nondecreasing. Hence, the answer is YES.
- **Test case 2:** The array is already balanced, so the answer is YES.
- **Test case 3:** No swaps allowed ($k = 0$) and the array is not balanced, so the answer is NO.
- **Test case 4:** Mirrored sums are $10 + 9 = 19$, $1 + 2 = 3$, which is not nondecreasing. Swapping 10 and 1 gives $[1, 10, 2, 9]$, sums $1 + 9 = 10$, $10 + 2 = 12$, which is balanced. Hence, YES.